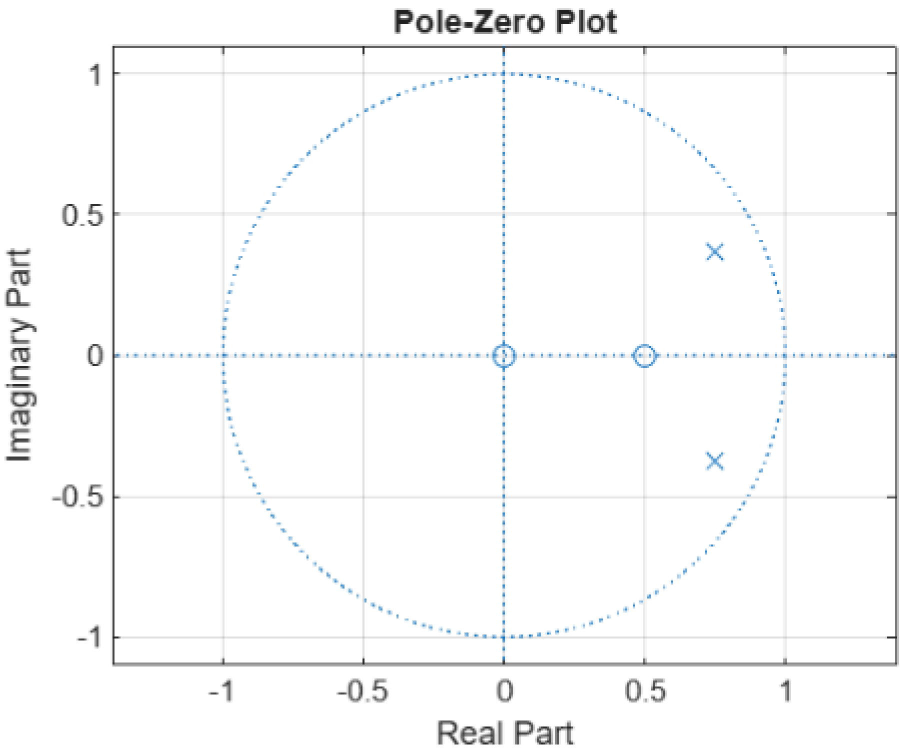
10. TRANSFORM TECHNIQUES AND SYSTEM REPRESENTATION

Experiment 10.1% Pole-Zero Plot

```
clc;
clear;
close all;
% Numerator coefficients
b = [1 -0.5];
% Denominator coefficients
a = [1 -1.5 0.7];
% Pole-Zero Plot
figure;
zplane(b,a);
grid on;
```

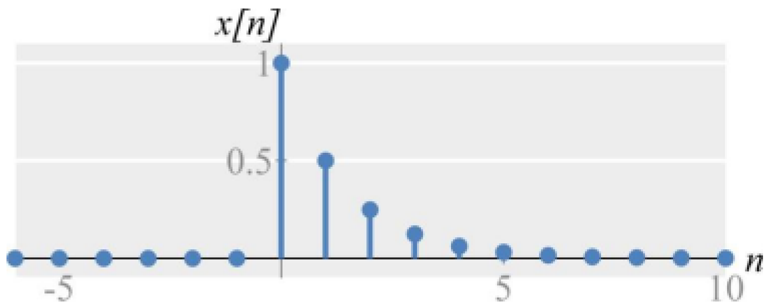
```
title('Pole-Zero Plot');
```



Experiment 10.2

```
% Z-Transform and Inverse Z-Transform
clc;
clear;
close all;
syms n z
% Signal
x = (0.5)^n;
% Z-Transform
X = ztrans(x,n,z);
disp('Z-Transform X(z) = ');
```

```
disp(X);
% Inverse Z-Transform
x_inv = iztrans(X,z,n);
disp('Inverse Z-Transform x(n) = ');
disp(x_inv);
```



Experiment 10.3

```
% Z-Transform of a Given Signal
clc;
clear;

close all;
syms n z
% Given signal
x = (0.8)^n;
% Z-Transform
X = ztrans(x,n,z);
disp('Z-Transform X(z) = ');
```

Z-Transform X(z) =
z/(z - 0.8)

Sample Values

n	x(n)
0	1.0000
1	0.8000
2	0.6400
3	0.5120

disp(X);

Experiment 10.4

% Laplace Transform of Continuous-Time Signal

clc;

clear;

close all;

syms t s

% Continuous-time signal

x = exp(-2*t);

% Laplace Transform

X = laplace(x,t,s);

```
disp('Laplace Transform X(s) = ');
disp(X);
```

Experiment 10.4% Laplace Transform of Continuous-Time Signal

```
clc;
clear;
close all;

syms t s
% Continuous-time signal
x = exp(-2*t);
% Laplace Transform
X = laplace(x,t,s);
disp('Laplace Transform X(s) = ');
```

```
Laplace Transform X(s) =
```

$$\frac{1}{s+2}$$

```
disp(X);
```